

Your Institution
Your Department

Friday 31st October, 2025

Admissions Committee / Hiring Manager

Dear Members of the Committee,

I am writing to provide a recommendation for [Candidate Name], whom I have supervised for approximately seven months on a research project in maritime radar-camera sensor fusion. I can speak to their technical capabilities, research approach, and development areas that would be valuable context for your evaluation.

Technical Strengths

[Candidate] demonstrates exceptional technical breadth across multiple domains: radar signal processing, computer vision, deep learning, and systems engineering. This is genuinely unusual at the Master's level. They have strong foundational knowledge of marine radar physics, CFAR detection algorithms, and neural network architectures. During our project, they implemented a working Constant False Alarm Rate detector on real radar data, designed a multi-sensor calibration framework, and coordinated hardware integration of radar and optical systems. Rather than purely theoretical work, they built actual systems and conducted data collection in the field.

Their ability to articulate complex technical problems is clear. The project's core motivation—using radar's all-weather detection to compensate for camera limitations in maritime environments—is well-reasoned and shows understanding of how to combine sensor strengths. Their literature review was comprehensive, synthesizing 131+ references across multiple fields into coherent problem framing.

Communication and Planning

[Candidate] communicates clearly through presentations and documentation. They consistently delivered monthly progress reports, maintained regular team meetings, and have begun mentoring undergraduate researchers. This indicates emerging leadership capacity and ability to articulate ideas to audiences with varying technical backgrounds.

Important Development Areas

I want to be transparent about areas where [Candidate] would benefit from continued growth, as this represents honest feedback rather than concerns about capability.

Integration and Completion Focus

The primary observation is that while individual components of the project are well-developed,

the end-to-end integration remains incomplete. Seven months into a 12-month project, the core system—radar detection and automated camera labeling—is not yet functional as a unified pipeline. This pattern suggests that [Candidate] may benefit from stronger emphasis on completing integrated systems before exploring additional technical directions.

During the project, parallel explorations of Vision Transformers, Masked Autoencoders, integer processing, and GPU optimization proceeded while the basic CNN-CFAR comparison remained unfinished. While this breadth demonstrates intellectual curiosity, it also suggests that prioritization between exploration and completion could be more disciplined. Successful research often requires resisting attractive tangents to finish foundational work first.

Quantitative Rigor and Validation

[Candidate] excels at conceptual development and theoretical motivation but would strengthen their research by earlier emphasis on quantitative validation. Claims about performance improvements are sometimes stated theoretically (“should reduce labeling time,” “expected to improve detection”) without corresponding metrics. In machine learning and signal processing, measurement is essential; approaches cannot be considered validated until compared against clear baselines with statistical rigor.

For example, while the multi-gain radar methodology is well-motivated, its actual benefit to detection performance has not been quantitatively demonstrated. Similarly, CNN model performance relative to the CFAR baseline has not been reported, despite being central to the project’s value proposition.

Problem-Solving Methodology

When encountering obstacles—such as sensor calibration challenges or clutter complexity—[Candidate] sometimes moves to alternative approaches without fully documenting diagnostic steps or recovery attempts. Strengthening systematic problem-solving documentation (hypothesis, test, result, analysis) would increase confidence in their troubleshooting methodology.

The sensor calibration challenge, identified as critical to the project’s success, is still unresolved in October. Understanding whether this reflects technical difficulty (expected) or less disciplined approach to hard problems (concerning) requires clearer documentation of attempted solutions.

Recommendation

I recommend [Candidate] for graduate programs and research positions, with the understanding that their strengths lie in technical breadth and problem identification. They would be most successful in environments that:

1. Emphasize project completion and integration as key milestones, not just technical exploration

2. Require quantitative validation before considering work “complete”
3. Provide structured feedback on prioritization and scope management
4. Value their ability to understand complex multi-domain problems
5. Leverage their communication skills for collaborative research

Their technical knowledge is strong and genuine. Their growth areas—focus, quantitative rigor, systematic problem-solving—are addressable through experience and mentorship and do not reflect fundamental limitations.

For a role requiring deep technical knowledge, broad learning ability, and problem identification, I recommend [Candidate] without reservation. For a role requiring execution of tightly-scoped, completion-focused work, I recommend them contingent on environmental factors that emphasize those skills.

They have clear potential for continued development as a researcher and engineer.

Sincerely,

Your Name